Unemployment Insurance Generosity and Enrollment into Postsecondary Education

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Abstract

This paper examines whether reductions in unemployment insurance (UI) generosity affect enrollment in postsecondary education. Between 2011 and 2016, nine U.S. states reduced the maximum duration of UI benefits, creating variation in access to unemployment compensation. Using institution-level enrollment data from the Integrated Postsecondary Education Data System (IPEDS) from 2007–2019 and a stacked difference-in-differences design, I estimate the effect of these policy changes on enrollment at postsecondary institutions. Results indicate that UI benefit reductions increased certificate program enrollment by approximately 5.8 percent, but had no measurable effect on degree-granting institutions. The findings suggest that reduced UI generosity may encourage displaced workers to invest in short-term occupational training, highlighting postsecondary certificate programs as an important educational response to changes in labor market policy.

1 Introduction

Unemployment insurance (UI) is one of the largest components of the United States social safety net, providing temporary income replacement to workers who lose their jobs through

no fault of their own. While the primary objective of UI is to smooth consumption during unemployment, a substantial literature shows that changes in benefit generosity also affect labor market behavior. Reductions in UI generosity shorten unemployment durations and UI receipt (Johnston and Mas 2016), may affect wages offered by employers (Dahl and Knepper 2022), and are associated with exits from the labor force (Card, Chetty, and Weber 2007; GAO 2015; Conway 2018). Comparatively little attention, however, has been given to whether changes in UI generosity affect investments in postsecondary education.

This paper examines whether reductions in unemployment insurance generosity affect enrollment in postsecondary education. The analysis exploits a series of state policy reforms implemented between 2011 and 2016, during which nine states reduced the maximum duration of regular unemployment insurance benefits below the long-standing 26-week standard. These staggered policy changes provide variation in UI generosity across states and over time.

To estimate the effect of these reforms, I construct an institution-year panel using enrollment data from the Integrated Postsecondary Education Data System (IPEDS) covering 2007–2019. I then merge state-level information on UI policy to each institution according to the state in which the institution is located. The empirical analysis employs the stacked difference-in-differences estimator developed by Wing, Freedman, and Hollingsworth (2024), which is designed for settings with staggered treatment timing.

The results indicate that reductions in UI generosity increased enrollment at non-degree-granting certificate institutions by approximately six percent. The pre-treatment event-study coefficients show little evidence of differential pre-trends, providing support for the parallel-trends identifying assumption. Enrollment gains emerge following implementation of the policy changes and persist for several years. In contrast, I find no comparable effects for community colleges or four-year institutions.

To understand why enrollment in postsecondary education may be a response to changes in UI, consider a discrete-time model of unemployment. In each period of an unemployment

spell, a worker may accept an available job offer, continue searching, or leave the labor force temporarily to pursue another activity, including education or training. The value of continued search depends partly on expected future UI benefits. Shortening the maximum duration of UI therefore lowers the value of continued search and causes some workers to reach benefit exhaustion sooner. Reducing UI therefore raises the relative value of leaving the labor force to reskill. Section 1.1 develops this intuition in greater detail.

Some institutions are likely to be more attractive to people looking to reskill and rejoin the labor force than others and certificate programs seem to be the most natural fit for poeple in this position. Postsecondary certificate programs have become an increasingly important component of workforce development in the United States. Unlike traditional degree programs, certificates generally require between several weeks and two years to complete and are designed to provide occupation-specific skills for relatively rapid labor market entry. These programs serve a population that differs substantially from traditional college students: certificate students tend to be older, are more likely to have prior labor market experience, and are more likely to have dependents. These features make certificate programs a particularly relevant margin on which workers experiencing unemployment may adjust.

This paper first contributes to the literature on unemployment insurance and worker responses to changes in benefit generosity. Existing work has focused primarily on outcomes such as unemployment duration, UI receipt, wages, and labor-force participation. Johnston and Mas (2016), for example, show that reductions in UI generosity shorten unemployment spells and reduce UI use, while Dahl and Knepper (2022) document effects on wages offered by firms. Other work, including Card, Chetty, and Weber (2007), the GAO (2015), and Conway (2018), documents labor-force exits associated with changes in UI. I complement this literature by examining postsecondary enrollment as a potential destination for workers whose behavior changes when UI becomes less generous.

Second, this paper contributes to the literature on postsecondary certificate programs and non-degree credentials. Existing work has largely documented who enrolls in certificate

programs, how students finance these programs, and the growing role of certificates in the labor market. Baum, Holzer, and Luetmer (2020) and Carnevale et al. (2020) describe certificate students and their financing, Crockett, Perlemeter, and Zhang (2024) document growth in certificate programs and employer demand for credentials, and Ositelu (2021) provides survey evidence on perceptions of their labor-market value. In contrast to this primarily descriptive literature, this paper studies whether a change in the economic incentives facing unemployed workers causally affects enrollment in these programs.

The remainder of the paper proceeds as follows. Section 1.1 presents the conceptual framework. Section 2 describes the data and empirical strategy. Section 3 presents the main results and robustness analysis. Section 4 discusses the findings and concludes.

1.1 Conceptual Framework

Consider an unemployed worker making decisions in discrete time. In each period, the worker may receive a job offer. If the offer is sufficiently attractive, the worker can accept employment. Otherwise, the worker chooses whether to continue searching or to leave the labor force temporarily, potentially to acquire new skills through postsecondary education. This framework allows reductions in UI generosity to increase transitions into employment and transitions into education on separate margins.

The value of continued search depends in part on the UI benefits the worker expects to receive in future periods. When an acceptable job offer does not arrive, the worker advances to the next period of unemployment with fewer remaining weeks of UI eligibility. As benefit exhaustion approaches, the value of continued search declines. Once benefits are exhausted, that value may fall substantially. A reduction in the maximum duration of UI accelerates this process: relative to an otherwise similar worker facing more generous benefits, a worker subject to a shorter maximum duration reaches benefit exhaustion after fewer periods without an acceptable job offer. Shortening benefit duration therefore increases the number of unemployed workers who reach a point at which continued search is less

attractive relative to other activities.

A decline in the value of continued search need not cause all affected workers to accept lower-paying employment. Reservation wages may adjust only modestly over an unemployment spell, and some workers may face fixed costs of employment that make substantially lower-paying offers unattractive. Parents facing childcare expenses are one example. Moreover, lowering a reservation wage does not itself ensure that a worker receives an acceptable offer. For workers who approach or reach benefit exhaustion without finding an attractive employment opportunity, temporarily leaving the labor force to acquire new skills should therefore become relatively more valuable.

This mechanism also suggests why the enrollment response may be concentrated in certificate programs. Certificate programs typically require substantially less time to complete than associate's or bachelor's degree programs and often provide occupation-specific training. They also disproportionately enroll older students and individuals with prior labor-market experience. A worker attempting to improve employment prospects while returning to work relatively quickly may therefore find a short certificate program more attractive than beginning a longer degree program. The empirical analysis consequently examines certificate-granting institutions, community colleges, and four-year institutions separately.

2 Methodology

2.1 Data

I construct an institution-year panel by combining institution-level enrollment data with state-level information on unemployment insurance policy. Annual enrollment comes from the Integrated Postsecondary Education Data System (IPEDS), which provides information on postsecondary institutions participating in federal student aid programs. The sample spans 2007 through 2019. I merge the state-level UI policy data to each institution according to the state in which the institution is located. Information on changes in UI policy comes

from the Congressional Research Service. Between 2011 and 2016, nine states reduced the maximum duration of regular UI benefits below the traditional 26-week maximum, generating staggered treatment timing across states.

The primary outcome is the natural logarithm of annual enrollment at institution i in year t. Because the mechanism described in Section 1.1 implies that workers seeking to return to employment relatively quickly may respond differently across types of postsecondary education, I perform the analysis separately for three groups of institutions: non-degree-granting institutions below the community-college level, community colleges, and institutions granting bachelor's degrees or higher. Certificate-granting institutions are of particular interest because their students are more likely to have prior labor-market experience and because certificate programs are generally shorter to complete than degree programs. These features may make certificate programs especially attractive to unemployed workers seeking to acquire new skills and return to work quickly rather than undertake a longer course of study.

2.2 Empirical Strategy

The empirical analysis employs the stacked difference-in-differences estimator proposed by Wing, Freedman, and Hollingsworth (2024), which is designed for settings with staggered treatment adoption. Conventional two-way fixed effects estimators can produce biased estimates when treatment effects vary over time or across cohorts. The stacked estimator avoids these issues by constructing separate event-study samples for each treatment cohort and comparing treated states only with states that are never treated.

For each treatment year, a separate sub-experiment is constructed consisting of all states adopting the policy in that year together with all never-treated states. Each sub-experiment is restricted to a balanced event window extending four years prior to treatment and five years following treatment. These cohort-specific panels are then vertically stacked to create the final estimation sample, where time is indexed relative to the treatment year rather than calendar year.

2.3 Event-Study Specification

The primary estimating equation is

$$y_{ice} = \alpha + \gamma D_{ic} + \sum_{\substack{h=-4 \\ h\neq -1}}^{5} (\lambda_h \mathbf{1}[e = h] + \delta_h D_{ic} \times \mathbf{1}[e = h]) + \varepsilon_{ice}, \tag{1}$$

where y_{ice} denotes the natural logarithm of annual enrollment for institution i in sub-experiment c at event time e. The indicator variable D_{ic} equals one for institutions located in states treated within sub-experiment c and zero for institutions located in never-treated states. The omitted event period is one year prior to treatment ($e = -1$), so each coefficient δ_h measures the difference in enrollment between treated and comparison institutions relative to the year immediately preceding treatment.

Because never-treated comparison states appear in multiple sub-experiments, the stacked dataset contains repeated observations for comparison units. Following Wing et al. (2024), weighted least squares is used to ensure that comparison observations receive appropriate weight across sub-experiments. Treated observations receive a weight of one, while comparison observations receive weights proportional to the relative number of treated and untreated units in each cohort-specific experiment. Formally, the weights are given by

$$Q_{ic} = \begin{cases} 1, & \text{if } D_{ic} = 1, \\ \dfrac{N_a^D / N^D}{N_a^C / N^C}, & \text{if } D_{ic} = 0, \end{cases} \tag{2}$$

where N_a^D denotes the number of treated units in sub-experiment a, N^D is the total number of treated units across all trimmed sub-experiments, N_a^C is the number of comparison units in sub-experiment a, and N^C is the total number of comparison units.

The event-study specification serves two purposes. First, estimates for the pre-treatment periods provide evidence relevant to the parallel-trends assumption by showing whether treated and comparison institutions exhibited similar enrollment trajectories prior to im-

plementation of the UI reforms. Second, the post-treatment coefficients characterize the dynamic effects of reductions in UI generosity over time. The average treatment effect on the treated (ATT) is calculated as the average of the post-treatment event-time coefficients.

Identification relies on the assumption that, absent changes in unemployment insurance generosity, enrollment trends in treated states would have evolved similarly to those in never-treated states. Under this parallel trends assumption, differences in enrollment following treatment can be interpreted as the causal effect of reductions in UI generosity.

Several features of the research design strengthen this interpretation. First, the analysis exploits policy variation generated by legislative changes to state unemployment insurance systems rather than changes in institutional behavior. Second, the stacked difference-in-differences estimator compares each treatment cohort only with never-treated states, eliminating the contamination that can arise when already-treated states serve as comparisons. Finally, the event-study estimates permit an examination of pre-treatment trends, which provides supporting evidence for the identifying assumption but does not itself test the parallel-trends assumption.

3 Results

This section presents the estimated effects of reductions in unemployment insurance (UI) generosity on enrollment in postsecondary institutions. As described in the data section, the analysis is conducted separately for certificate-granting institutions, community colleges, and four-year and post-baccalaureate institutions. I begin with the certificate-granting institutions and then report corresponding estimates for community colleges and four-year institutions. The pre-treatment coefficients are used as descriptive evidence on pre-trends rather than as a formal test of the parallel-trends assumption.

Table 1 reports covariate balance between the treated and comparison states on four dimensions that are likely to affect adoption of a restricted UI policy: GDP, the unemploy-

Covariate Balance

	Comparison	Treated	Difference	p-value
GDP	315089.36	339011.94	-23922.58	0.871
StateUnemploymentRate	8.45	9.97	-1.52	0.043
UnionMembership	0.12	0.07	0.05	0.014
divided_gov	0.56	0.33	0.23	0.224

Table 1: Values are for 2010

ment rate, union membership, and an indicator for divided state government. There is a potential concern that treated states are meaningfully different from comparison states in this analysis. As a robustness check, I later use a restricted sample of states that are more similar along these dimensions.

Table 2 summarizes the average treatment effects on the treated (ATTs) from the event-study specifications. Reporting the aggregate estimates separately makes the magnitude and precision of the results readily comparable across institution types.

Table 2: Average Treatment Effects on Enrollment by Institution Type

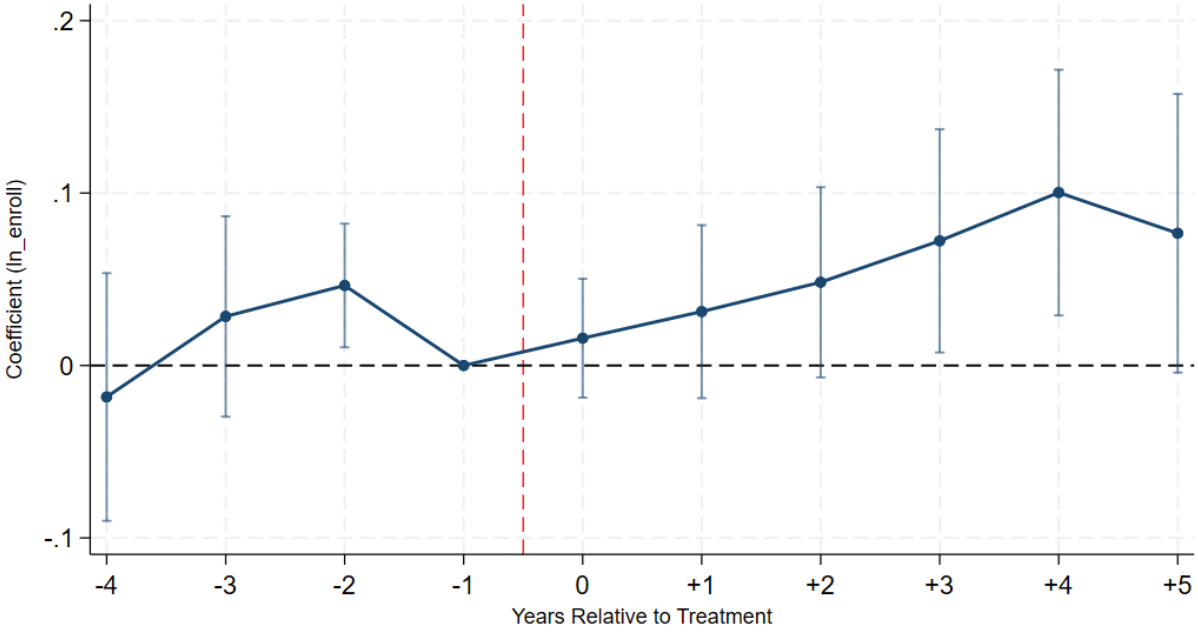
Institution type	ATT	Standard error
Certificate-granting institutions	0.0576	0.0340
Community colleges	-0.0038	0.0220
Four-year and post-baccalaureate institutions	-0.0132	0.0234

Notes: Entries report aggregate ATTs from the stacked event-study specifications. Standard errors are the corresponding aggregate standard errors. The outcome is log annual enrollment.

Figure 1 presents the event-study estimates for non-degree-granting institutions below the community college level. The omitted reference period is the year immediately preceding implementation of the UI reform ($t = -1$). Each coefficient therefore measures the treated–comparison difference in log enrollment at event time h relative to the difference in the year immediately preceding treatment.

The event-study estimates provide little evidence of differential pre-treatment trends between treated and comparison states. Coefficients during the four years preceding treatment remain close to zero and fluctuate without any systematic pattern, providing supporting

Figure 1: Event-Study Estimates of the Effect of Shorter Unemployment Insurance Duration on Enrollment — Certificate-Granting Institutions.



Notes: Estimates use the stacked difference-in-differences design of Wing, Freedman, and Hollingsworth (2024). The omitted period is event time −1. Error bars denote 95 percent confidence intervals.

evidence for the parallel-trends assumption underlying the stacked difference-in-differences design.

Following implementation of the UI reforms, enrollment increases relative to the comparison group. The estimated treatment effects become positive shortly after treatment and remain elevated throughout the post-treatment period. The absence of an immediate decline followed by sustained positive effects suggests that institutions experienced a persistent increase in enrollment rather than a temporary shift in the timing of enrollment decisions.

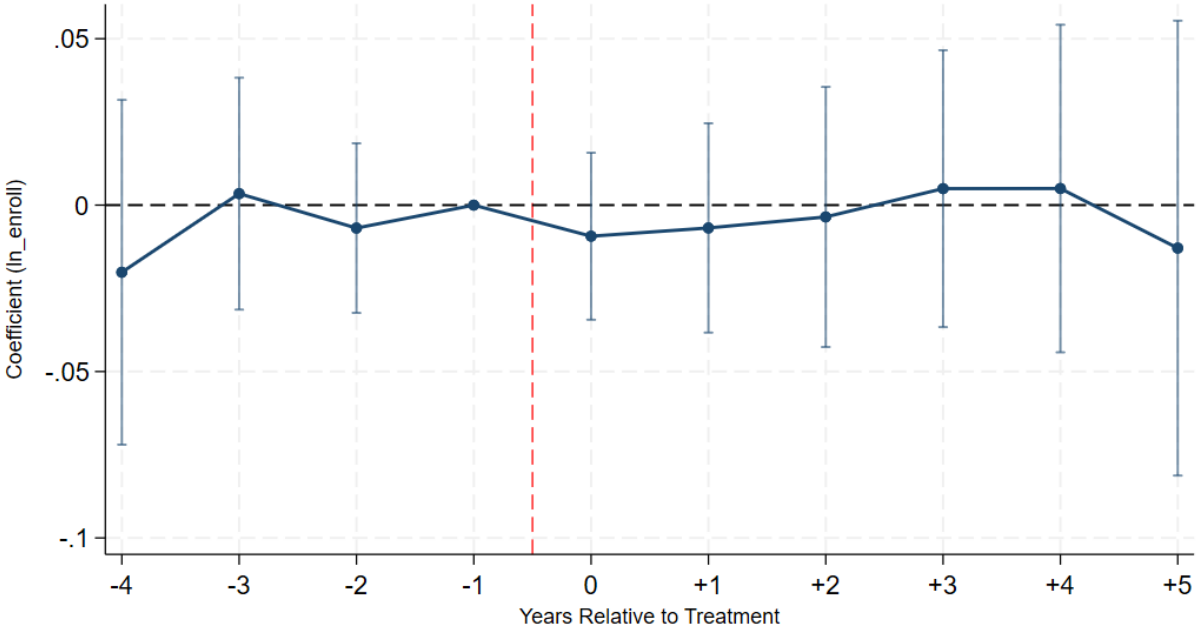
Averaging the post-treatment coefficients yields an estimated average treatment effect on the treated (ATT) of approximately 0.058 log points, implying that reductions in UI generosity increased enrollment at certificate institutions by roughly 5.8 percent. This magnitude is economically meaningful given the relatively modest size of the policy intervention and is consistent with the hypothesis that displaced workers respond to reduced unemployment benefits by investing in short-term occupational training.

The estimated treatment effects increase gradually following implementation of the UI reforms rather than appearing immediately. This pattern is consistent with the institutional features of certificate programs. Individuals must first become unemployed, adjust to the reduction in available UI benefits, research educational opportunities, complete the admissions process, and begin coursework. Because many certificate programs operate on fixed academic calendars, enrollment responses may not occur until the next available start date.

The persistence of the estimated effects through the fifth post-treatment year suggests that the reforms generated a sustained increase in demand for occupational training rather than merely shifting enrollment across academic terms. This finding is consistent with a behavioral response in which reduced income support permanently changes the relative attractiveness of human capital investment.

To examine whether the estimated effects are unique to certificate programs, I replicate the empirical analysis using institutions that primarily grant associate's degrees and institutions offering bachelor's degrees or higher. If reductions in UI generosity broadly increased

Figure 2: Event-Study Estimates of the Effect of Shorter Unemployment Insurance Duration on Enrollment — Community Colleges.



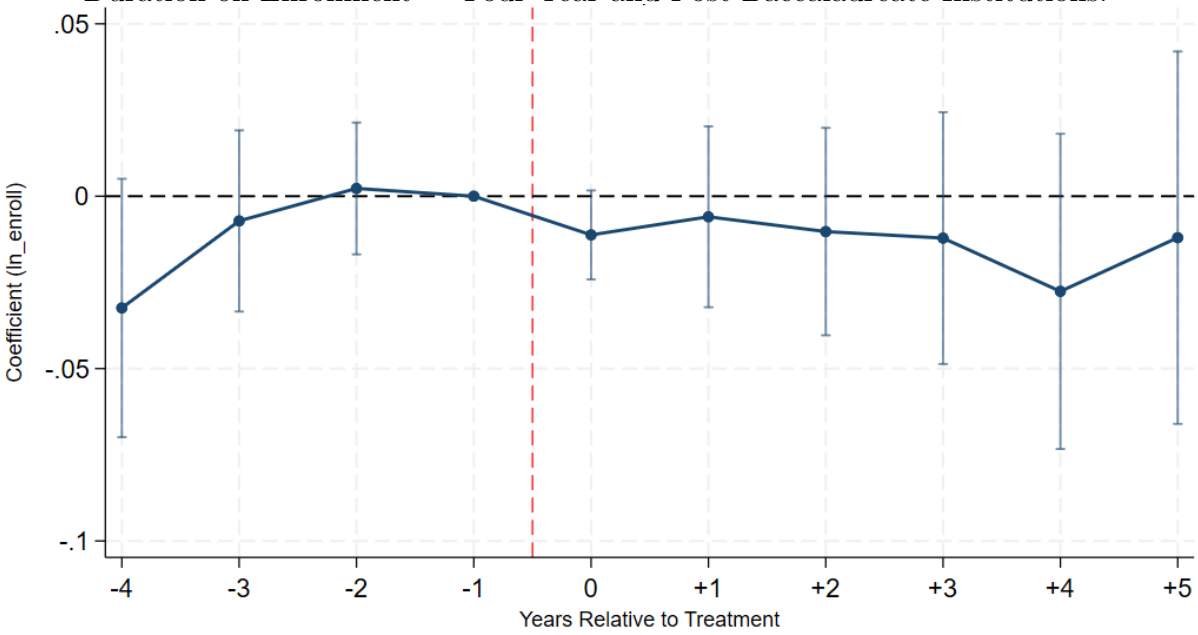
Notes: Estimates use the stacked difference-in-differences design of Wing, Freedman, and Hollingsworth (2024). The omitted period is event time −1. Error bars denote 95 percent confidence intervals.

demand for all forms of postsecondary education, similar effects would be expected across these sectors.

Figure 2 reports the event-study estimates for community colleges.

Unlike the results for certificate institutions, community colleges exhibit no sustained increase in enrollment following reductions in UI generosity. The estimated average treatment effect is approximately −0.004 log points, and the post-treatment coefficients fluctuate around zero without exhibiting a consistent upward trend. Although several estimates are imprecisely estimated, the overall pattern provides little evidence that reductions in UI benefits increased enrollment at community colleges. Community colleges share many of the features of certificate-only institutions that are appealing to people who have previously been in the work force, and I would expect the changes in UI to have some effect on community college enrollment. However, during the time period in this analysis people above the age of 23 were a minority of the people enrolled in community colleges. This study may not

Figure 3: Event-Study Estimates of the Effect of Shorter Unemployment Insurance Duration on Enrollment — Four-Year and Post-Baccalaureate Institutions.



Notes: Estimates use the stacked difference-in-differences design of Wing, Freedman, and Hollingsworth (2024). The omitted period is event time −1. Error bars denote 95 percent confidence intervals.

be sufficiantly powered to detect the effects of UI changes in this setting.

Figure 3 presents analogous estimates for four-year colleges and universities.

The estimated effects for four-year institutions are likewise small and centered near zero. The average treatment effect of approximately −0.019 log points suggests that reductions in UI generosity did not meaningfully affect enrollment at bachelor's degree-granting institutions.

Taken together, these results suggest that reducing UI meaningfully increases enrollments in postsecondary education, but the effects are concentrated in institutions that primarily older individuals that are more likely to have been previously employed. Rather than increasing enrollment across all sectors of higher education, reductions in UI generosity appear to have affected only institutions offering relatively short-term occupational credentials. This pattern is consistent with the proposed mechanism that unemployed workers respond to lower UI generosity by acquiring labor market skills that can be obtained quickly and facilitate

rapid re-entry into employment.

3.1 Robustness to Alternative Comparison Group Selection

A potential concern with the baseline specification is that states that reduced the maximum duration of unemployment insurance benefits may differ systematically from states that did not adopt these reforms. In particular, differences in economic conditions or state political environments could be correlated with both the adoption of UI reforms and subsequent changes in postsecondary enrollment. I therefore examine the robustness of the main results to restricting the comparison group to states that are more similar to the treated states along a set of predetermined characteristics.

I construct the restricted comparison group using four state-level characteristics measured in 2010, prior to the first UI reforms in the sample: GDP, the unemployment rate, union membership, and an indicator for divided state government. Starting with the full set of never-treated states, I sequentially remove 15 comparison states. At each iteration, I calculate the imbalance between the treated and remaining comparison states as the sum of the squared differences in the standardized means of the four covariates. I then remove the comparison state whose exclusion produces the smallest resulting imbalance. Repeating this procedure 15 times produces a restricted comparison sample selected without reference to post-treatment enrollment outcomes.

Table 3 reports covariate balance for the resulting sample. The restricted comparison group is substantially more similar to the treated states along the dimensions that exhibited the greatest imbalance in the unrestricted sample. In particular, the difference in the 2010 unemployment rate falls from 1.52 percentage points in the unrestricted sample to 0.99 percentage points in the restricted sample, while the difference in union membership falls from 0.05 to 0.03. None of the four differences in the restricted sample is statistically significant at the five-percent level. The restricted sample excludes Alaska, Connecticut, Hawaii, Iowa, Minnesota, Montana, Nebraska, New Jersey, New York, North Dakota, Ohio,

Pennsylvania, Vermont, Washington, and Wyoming.

Covariate Balance

	Comparison	Treated	Difference	p-value
GDP	314492.58	339011.94	-24519.36	0.879
StateUnemploymentRate	8.97	9.97	-0.99	0.186
UnionMembership	0.10	0.07	0.03	0.087
divided_gov	0.46	0.33	0.13	0.517

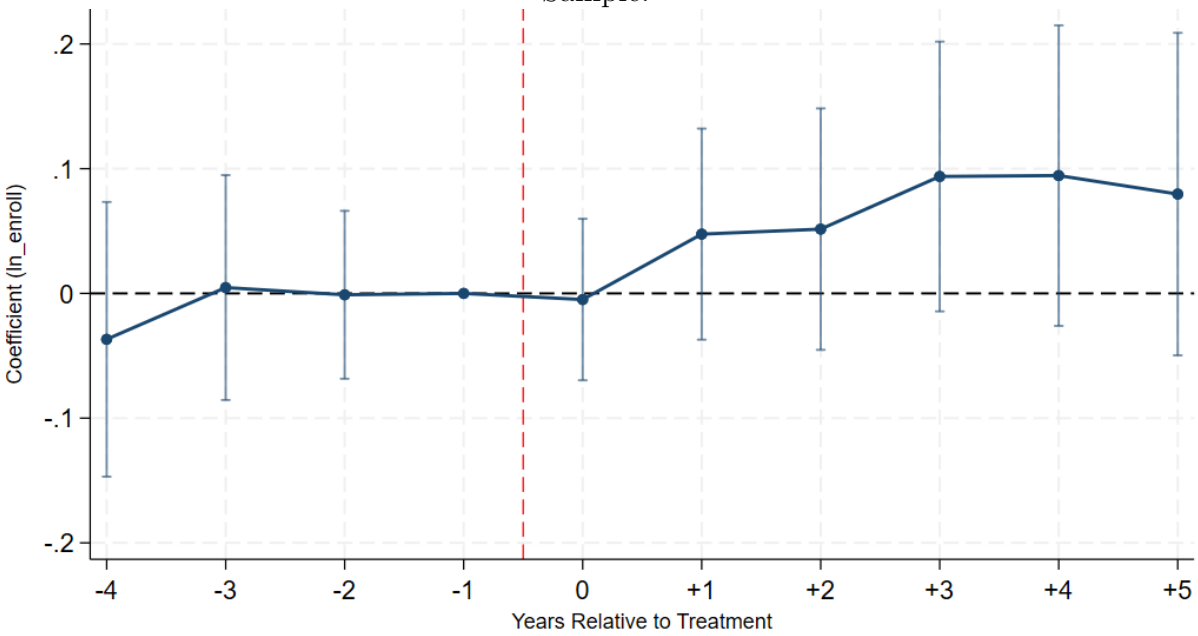
Table 3: Values are for 2010. Includes all states except: ALASKA, CONNECTICUT, HAWAII, IOWA, MINNESOTA, MONTANA, NEBRASKA, NEW JERSEY, NEW YORK, NORTH DAKOTA, OHIO, PENNSYLVANIA, VERMONT, WASHINGTON, and WYOMING.

I then re-estimate the stacked difference-in-differences specification using the restricted comparison sample. Figure 4 presents the event-study estimates for certificate-granting institutions. The results are qualitatively similar to the baseline estimates. The pre-treatment coefficients are centered near zero, while enrollment increases following the reduction in UI generosity. The post-treatment coefficients become increasingly positive over the first several years following treatment, with the largest point estimates occurring approximately three to five years after the reform.

The corresponding analyses for degree-granting institutions provide a useful comparison. Figures 5 and 6 report estimates for community colleges and four-year institutions, respectively. In contrast to certificate-granting institutions, neither group exhibits a persistent post-treatment increase in enrollment. The community-college estimates fluctuate around zero following treatment, while the estimates for four-year institutions are likewise generally small and show no systematic post-treatment pattern.

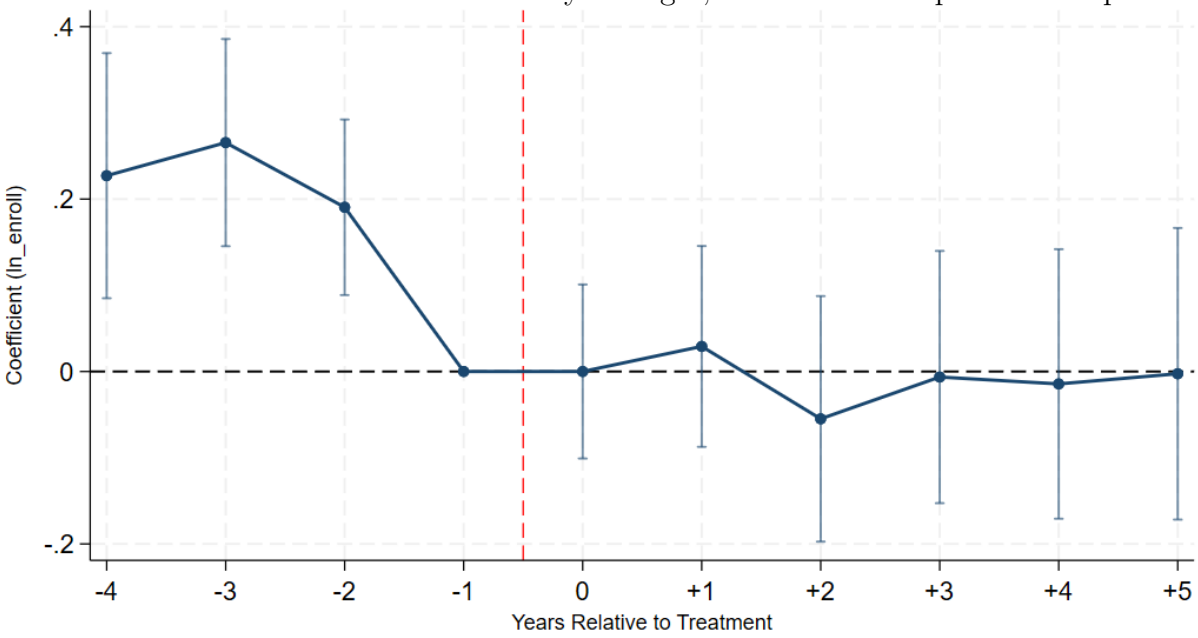
Taken together, these results indicate that the main finding is not driven solely by differences in observable economic or political characteristics between states that did and did not reduce UI generosity. Restricting the comparison group to states that are more similar to the treated states in terms of pre-reform GDP, unemployment, union membership, and divided government preserves the central empirical pattern: reductions in UI generosity are followed

Figure 4: Event-Study Estimates of the Effect of Shorter Unemployment Insurance Duration on Enrollment — Certificate-Granting Institutions, Restricted Comparison Sample.



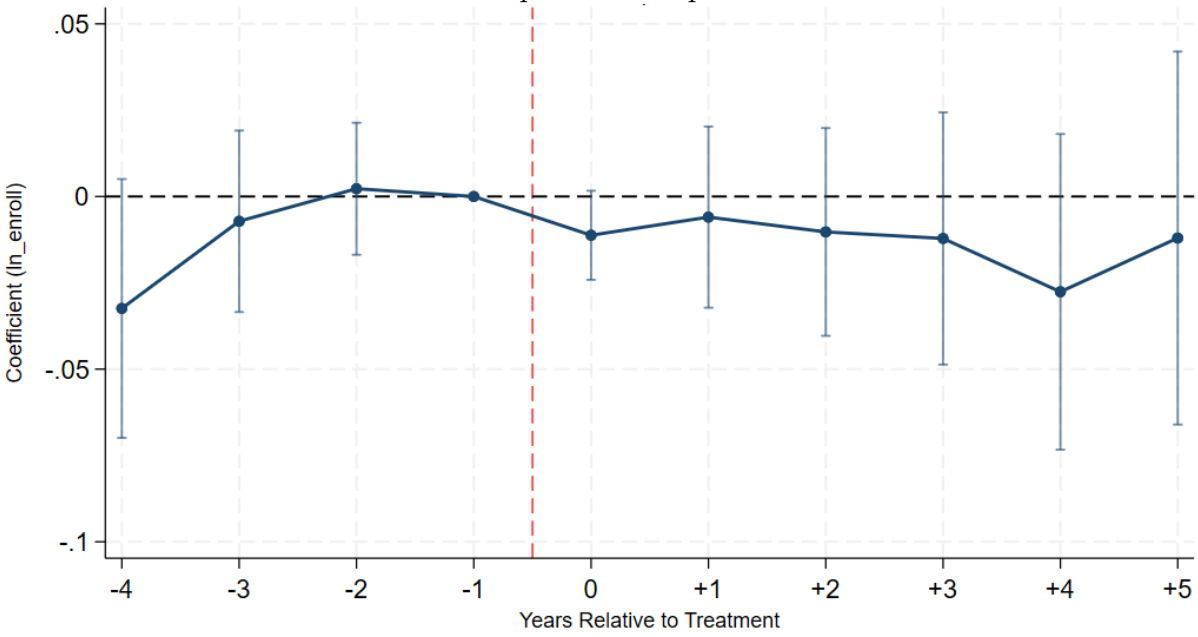
Notes: Estimates use the stacked difference-in-differences design of Wing, Freedman, and Hollingsworth (2024). The omitted period is event time −1. Error bars denote 95 percent confidence intervals.

Figure 5: Event-Study Estimates of the Effect of Shorter Unemployment Insurance Duration on Enrollment — Community Colleges, Restricted Comparison Sample.



Notes: Estimates use the stacked difference-in-differences design of Wing, Freedman, and Hollingsworth (2024). The omitted period is event time −1. Error bars denote 95 percent confidence intervals.

Figure 6: Event-Study Estimates of the Effect of Shorter Unemployment Insurance Duration on Enrollment — Four-Year and Post-Baccalaureate Institutions, Restricted Comparison Sample.



Notes: Estimates use the stacked difference-in-differences design of Wing, Freedman, and Hollingsworth (2024). The omitted period is event time −1. Error bars denote 95 percent confidence intervals.

by increased enrollment at certificate-granting institutions, with no analogous increase at community colleges or four-year institutions. This exercise therefore provides additional evidence that the estimated certificate-enrollment response is robust to alternative choices of the comparison group.

4 Discussion

The estimated increase in certificate enrollment is consistent with the theoretical framework presented earlier in the paper. Reductions in UI generosity decrease the opportunity cost of enrolling in postsecondary education by lowering the value of continued job search while simultaneously increasing the relative value of acquiring new occupational skills. The absence of comparable effects for community colleges and four-year institutions suggests that workers primarily respond by pursuing educational investments with relatively low completion times and direct labor market applications.

These findings also complement previous research documenting that reductions in UI generosity shorten unemployment durations and increase exits from the labor force. The present results suggest that at least part of these labor force exits may reflect transitions into postsecondary education rather than permanent withdrawal from economic activity. Consequently, evaluations of unemployment insurance policy should consider educational investment as an additional margin of adjustment when assessing the broader consequences of changes in benefit generosity.

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